Erfan Hosseini

Process and Kinetic Modeling | Techno-Economic Analysis | Life-Cycle Assessment

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Professional Summary

Chemical Engineering Ph.D. candidate specializing in process and kinetic modeling, techno-economic analysis (TEA), and life-cycle assessment (LCA) of bioenergy and waste-conversion systems. Experienced in Python-based dynamic simulation, kinetic parameter estimation, and automated scenario analysis, alongside process design using SuperPro Designer and Aspen Plus. Research combines process simulation with economic and environmental assessment, Monte Carlo uncertainty analysis, and sensitivity analysis to evaluate technology performance and identify conditions for viability. Additional experience includes machine-learning projects using scikit-learn.

Education

Auburn University — Auburn, Alabama

Ph.D. Candidate, Chemical Engineering; GPA: 3.6/4.0 — *Expected Summer 2027*

Advisor: Dr. Selen Cremaschi

Research focus: Computational modeling, techno-economic analysis, and life-cycle assessment of biochemical waste-conversion and bioenergy systems.

Relevant coursework: Advanced Biochemical Engineering; Linear Programming (AMPL); Stochastic Optimization and Risk; Big Data and Machine Learning for the Process Industry (Python); Probability and Statistics for Data Science (R).

Sharif University of Technology — Tehran, Iran

B.S., Chemical Engineering — *2021*

Research Experience

Graduate Research Assistant — Auburn University

Department of Chemical Engineering — *January 2022–Present*

Kinetic Modeling and Computational Analysis

- Extended a Python-based dynamic model of anaerobic digestion for pulp and paper sludge and food waste, incorporating biochemical reaction kinetics, component mass balances, gas–liquid transfer, physicochemical equilibria, and biological inhibition.
- Estimated feedstock-specific kinetic parameters from published experimental data using bounded nonlinear least squares and multistart initialization; achieved calibration R^2 values of 0.990 and 0.994.
- Developed automated Python workflows using NumPy, SciPy, and pandas for numerical simulation, convergence checking, data processing, scenario analysis, and visualization.
- Evaluated approximately 1,500 feed-composition cases across 15 food-waste categories to assess the effects of feed composition, loading rate, and retention time on methane productivity, conversion, and reactor stability.
- Identified feed mixtures with a model-predicted 17% higher maximum stable tested loading rate than the food-waste-only reference case.

Techno-Economic and Life-Cycle Assessment

- Developed a process simulation-based TEA/LCA framework for a hypothetical kraft mill processing 500 t/d of pulp and paper mill sludge; compared landfilling with baseline anaerobic digestion, alkaline pretreatment, and food-waste co-digestion under CHP and biomethane-upgrading pathways.
- Estimated capital and operating costs and evaluated net present value, return on investment, discounted payback, and minimum electricity and biomethane selling prices using discounted cash-flow analysis.
- Conducted comparative LCA in openLCA using ecoinvent 3.10 and nine TRACI 2.1 impact categories; developed process inventories and applied system expansion to account for displaced energy and avoided food-waste disposal.
- Modeled landfill methane using the IPCC first-order decay method and evaluated the effects of plant methane losses, landfill gas capture, and digestate management on climate impacts.
- Performed one-at-a-time sensitivity analysis and Monte Carlo uncertainty analysis with 10,000 draws and triangular input distributions to assess economic and environmental outcome ranges and the robustness of scenario comparisons.
- Evaluated plant size, energy prices, NaOH dose, co-digestion synergy, existing digester infrastructure, renewable fuel credits, and alternative digestate management to identify conditions affecting technology viability.
- Identified co-digestion as the lowest-cost PPMS digestion option per tonne of solids; demonstrated that all PPMS digestion options remained more costly than landfilling, while lowering global warming impacts, across all 10,000 Monte Carlo draws.

Process Simulation and Design

- Built commercial-scale flowsheets in SuperPro Designer for feed preparation, alkaline pretreatment, anaerobic digestion, solids separation, biogas cleaning, CHP generation, and biomethane upgrading.
- Completed mass and energy balances, equipment sizing, and utility-demand estimation using feedstock characterization, experimental results, and literature-based process assumptions.
- Developed Aspen Plus simulations of anaerobic digestion and CHP systems, incorporating estimated physical properties for biomass-derived components.
- Compared open-cycle and combined-cycle gas-turbine configurations and quantified utility consumption, heat recovery, and net power production to support conceptual process design.

Selected Publications and Research Dissemination

- **E. Hosseini**, Z. Jiang, and S. Cremaschi, "Opportunities and Challenges in Valorizing Pulp and Paper Sludge for Biogas Production via Anaerobic Digestion: A Review," *Sustainability Science and Technology*, 2025.
- **E. Hosseini**, S. Cremaschi, and Z. Jiang, "Techno-Economic Analysis of Methane Production from Pulp and Paper Sludge," *Systems and Control Transactions*, vol. 3, pp. 750–756, 2024.
- **E. Hosseini**, S. Cremaschi, and Z. Jiang, "Simulation-Based Screening of Anaerobic Digestion Strategies for Pulp and Paper Mill Sludge: Techno-Economic and Life Cycle Assessment under Combined Heat and Power and Biomethane Upgrading Pathways," manuscript under revision, *Bioresource Technology Reports*.
- Presented research on process modeling, bioenergy systems, and techno-economic assessment to academic and industry audiences at AIChE, FOCAPD, and pulp and paper industry conferences.

Technical Skills

Programming and Computational Modeling: Python (NumPy, SciPy, pandas, Matplotlib), first-principles modeling, dynamic simulation, biochemical reaction kinetics, numerical integration of ODEs, automated simulation workflows, data processing, and visualization.

Parameter Estimation and Model Evaluation: Bounded nonlinear least squares, multistart estimation, kinetic parameter estimation, model calibration, residual analysis, and regression.

Process Simulation and Design: Aspen Plus, SuperPro Designer, process flowsheet development, mass and energy balances, equipment sizing, utility-demand estimation, physical-property estimation, heat recovery, combined heat and power, and biomethane upgrading.

Techno-Economic Analysis: CAPEX and OPEX estimation, equipment cost scaling, discounted cash-flow analysis, net present value, minimum selling price, return on investment, discounted payback, and economic feasibility assessment.

Life-Cycle Assessment: openLCA, ecoinvent 3.10, TRACI 2.1, life-cycle inventory development, system-boundary and functional-unit definition, system expansion, landfill methane modeling, and environmental-impact interpretation.

Uncertainty and Sensitivity Analysis: Monte Carlo simulation, one-at-a-time sensitivity analysis, scenario comparison, and interpretation of economic and environmental uncertainty.

Optimization and Machine Learning: AMPL, linear programming, principal-component analysis (PCA), support-vector machines (SVM), and cross-validation; graduate training in stochastic optimization and decision-making under uncertainty.

Teaching, Mentoring, and Leadership Experience

Graduate Teaching Assistant — Auburn University
Digital Process Control, Thermodynamics, Fluid Mechanics, and Heat Transfer — *2022–2024*

- Guided students through engineering concepts and quantitative problem solving, evaluated technical work, and provided constructive feedback to students with varied academic and technical backgrounds.

Biomanufacturing Workshop Co-Leader — Alabama State University
Montgomery, Alabama — *February 2024*

- Co-developed and delivered a technical workshop for undergraduate research students on data analytics and machine-learning applications in industrial biomanufacturing.

Graduate Student Ambassador — Auburn University
Graduate-program outreach and prospective-student engagement — *2025–2026*

- Represented the graduate program in outreach activities and communicated research, academic, and professional-development opportunities to prospective students.

Selected Computational and Data-Science Projects

- **Optimization:** Formulated and solved linear-programming and network-flow problems using AMPL and completed graduate projects in stochastic optimization, scenario analysis, and decision-making under uncertainty.
- **Statistical learning:** Implemented principal-component analysis and support-vector-machine classification in Python for a team project using a Parkinson's disease dataset; reduced 22 variables to 8 principal components retaining 95% of variance and tuned model hyperparameters using five-fold cross-validation.
- **Image-based modeling:** Adapted convolutional-neural-network workflows in TensorFlow/Keras, including a multi-input model combining microscopy images with numerical experimental variables; performed data preprocessing, model training, and classification-performance evaluation.